FORECASTING FLORIDA HOME VALUES

A time-aware comparison of regression and tree-based models using monthly Zillow Home Value Index data

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Springboard Data Science Career Track | Capstone Two

Data snapshot: January 2000-January 2026 | Final model: Ridge regression

Executive summary

This project tests whether recent Florida home-value history can predict the next monthly Zillow Home Value Index (ZHVI). The original proposal aimed to identify undervalued U.S. markets, but the available dataset contains only state-level home values. I narrowed the final scope so the conclusions match the evidence.

Ridge regression delivered the lowest expanding-window validation MAE. On the independent January 2021-January 2026 holdout period, it achieved an MAE of $886, or 0.25%. That is a 67.6% reduction in MAE from predicting that next month will equal the prior month.

$886	$1,268	0.25%	0.999
Holdout MAE	Holdout RMSE	Holdout MAPE	Holdout R-squared

Recommendations

- Use the ridge model as a lightweight one-month monitoring forecast and retrain it when Zillow releases new monthly data.
- Do not use the tree models on the raw rising level series. They failed to extrapolate beyond values seen during training.
- Add rent, income, inventory, migration, mortgage-rate, and metro-level data before making an investment or undervaluation claim.

Problem and scope

Real estate investors need forward-looking information, but a useful investment ranking requires both expected growth and measures of value, income, and risk. This dataset supports the growth-monitoring part of that problem. The final modeling question is: **How accurately can information available through month t-1 predict Florida ZHVI in month t?**

Target and business use

- Target: Florida's monthly ZHVI in dollars.
- Forecast horizon: one month ahead.
- Potential use: short-term planning, portfolio monitoring, and a baseline for a larger housing model.

Data

The repository contains 313 monthly observations from January 2000 through January 2026 for 50 states and the District of Columbia. Florida has no missing values. Several other states begin later, so the revised workflow selects Florida before dropping missing rows. This preserves all valid Florida history.

The earlier notebook recorded Kaggle as the download location but did not preserve the exact page. Zillow is the underlying publisher. Zillow describes ZHVI as a measure of the typical home value and market changes in a region. ZHVI should not be interpreted as an individual property's sale price.

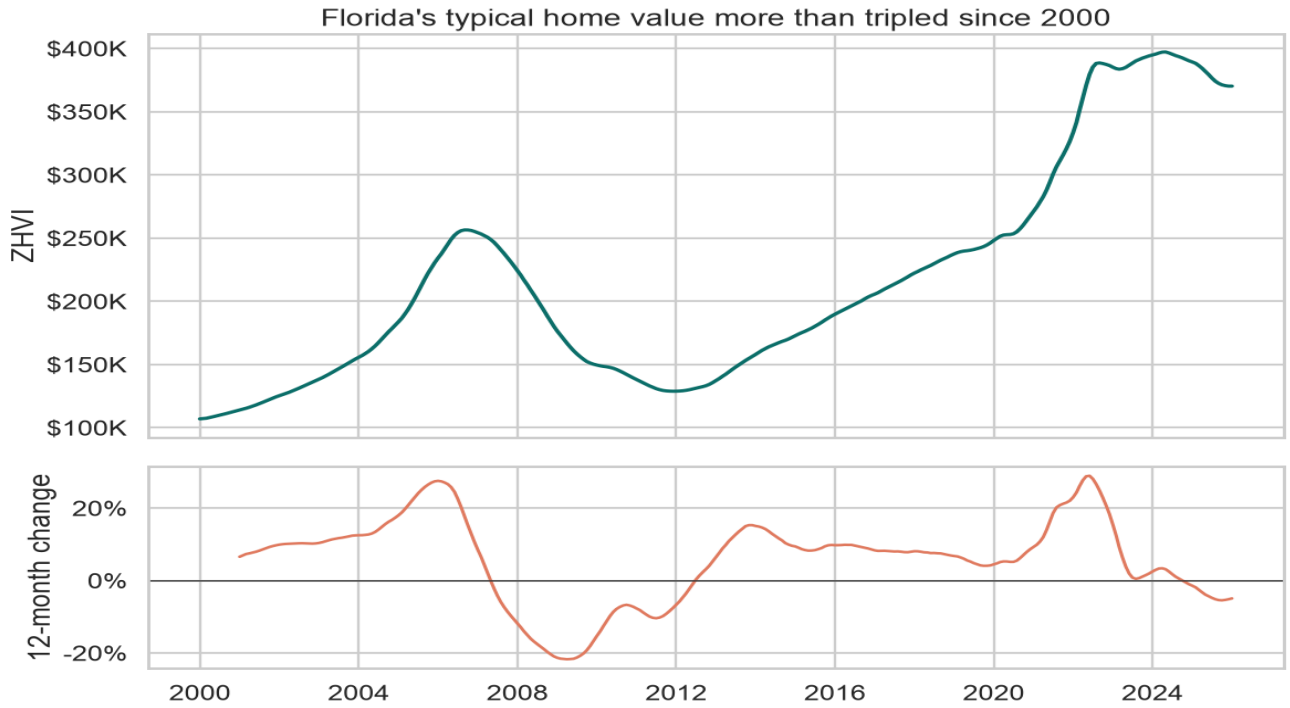


Figure 1. Florida ZHVI increased substantially over the sample, with especially rapid growth after 2020 followed by cooling.

Exploratory findings

The state comparison shows a broad upward pattern, but with different levels and timing. California remained the highest-value state in the four-state comparison. Florida and Texas began near similar values, then Florida moved higher after 2020.

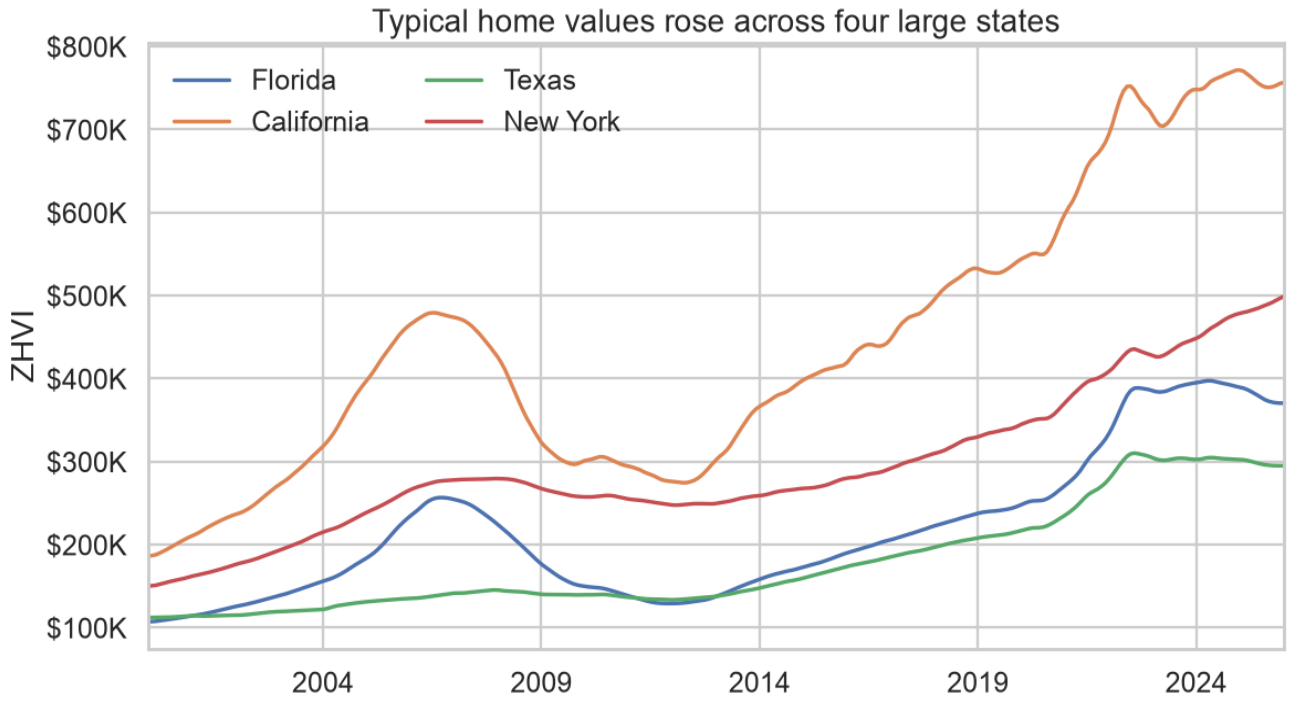


Figure 2. Monthly ZHVI for Florida, California, Texas, and New York.

The common direction across these markets suggests national forces matter, but a correlation between state price levels does not establish causation. Interest rates, supply, income, and population change are plausible drivers that this file does not contain.

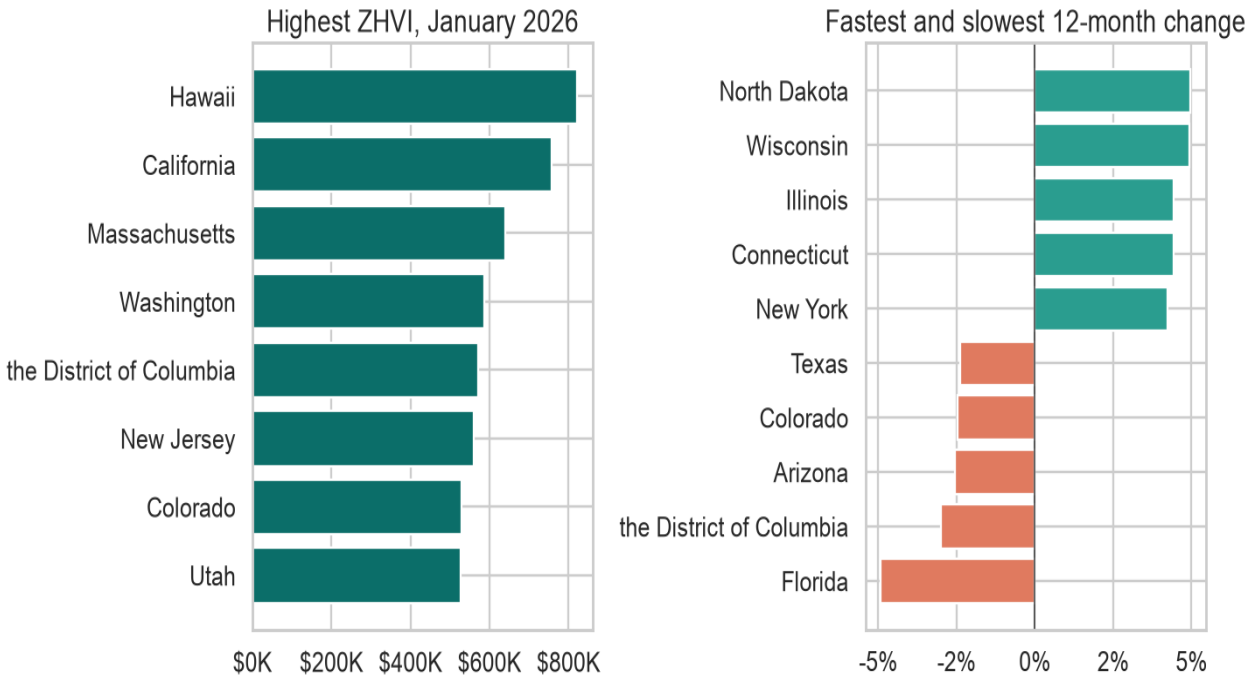


Figure 3. January 2026 home-value levels and recent one-year changes differ across states.

Modeling method

The revised model uses only information available before the month being predicted. Features include lags at 1, 2, 3, 6, and 12 months, rolling averages based on prior months, recent six-month variability, and cyclical month indicators.

Leakage prevention

- The first 80% of modeling rows form the training period; the latest 20% remain as a final holdout.
- Five-fold expanding-window validation preserves chronological order during model selection.
- Scaling occurs inside each regression pipeline, so every validation fold learns its scaler only from that fold's training data.

Models compared

- Persistence baseline: next month equals the prior month.
- Linear regression: an unregularized trend model.
- Ridge regression: a regularized linear model tuned over five alpha values.
- Random forest and gradient boosting: nonlinear tree ensembles tuned with the same time-aware folds.

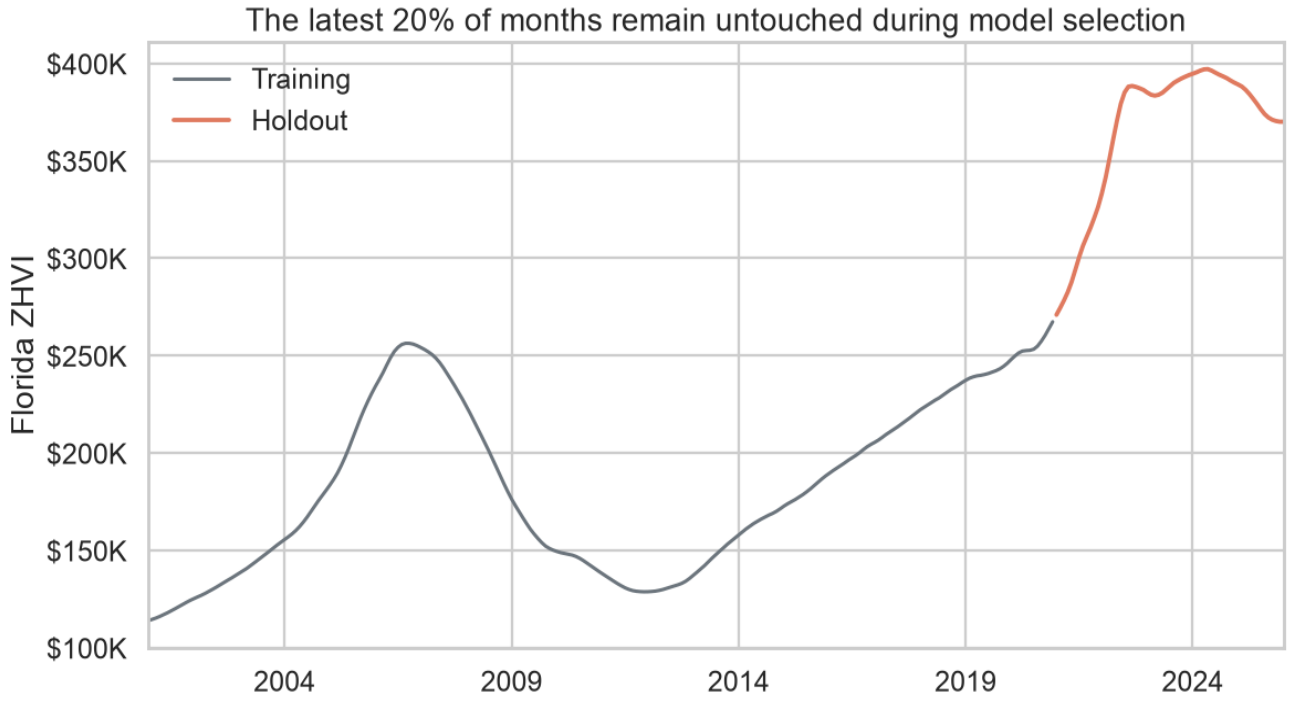


Figure 4. Training spans January 2001-December 2020; the holdout spans January 2021-January 2026.

Results

Ridge regression had the lowest validation error and was selected before the holdout results were reviewed. Linear regression happened to score lower error on the holdout. Keeping ridge as the final model protects the holdout from becoming another tuning set.

Model	CV MAE	Test MAE	Test RMSE	Test MAPE
Ridge regression	$1,292	$886	$1,268	0.25%
Persistence baseline	$1,812	$2,733	$3,813	0.80%
Linear regression	$2,680	$376	$486	0.10%
Gradient boosting	$17,433	$102,536	$109,044	27.13%
Random forest	$17,944	$101,385	$107,966	26.81%

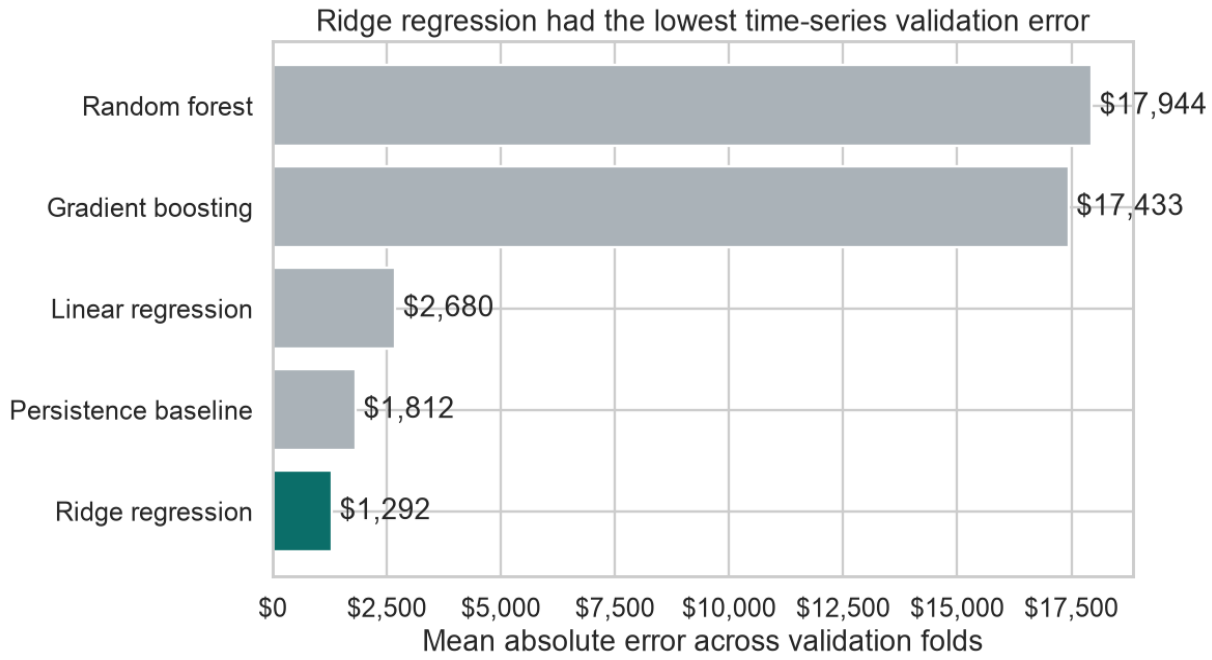


Figure 5. Time-series validation, rather than the holdout period, determines the final model.

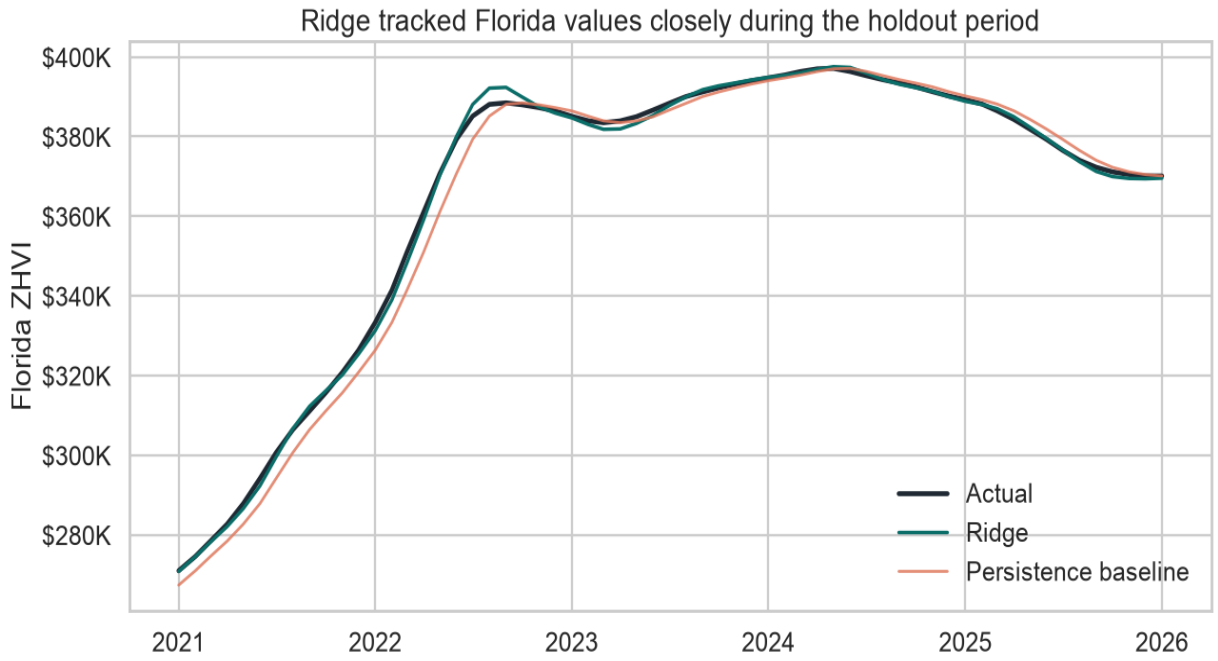


Figure 6. Ridge predictions track the holdout series closely and improve on the persistence baseline.

Interpretation and limitations

The ridge model's holdout error is small relative to Florida's home-value level. Its mean residual is positive, which indicates slight underprediction on average, and its largest absolute monthly residual is about $4,075.

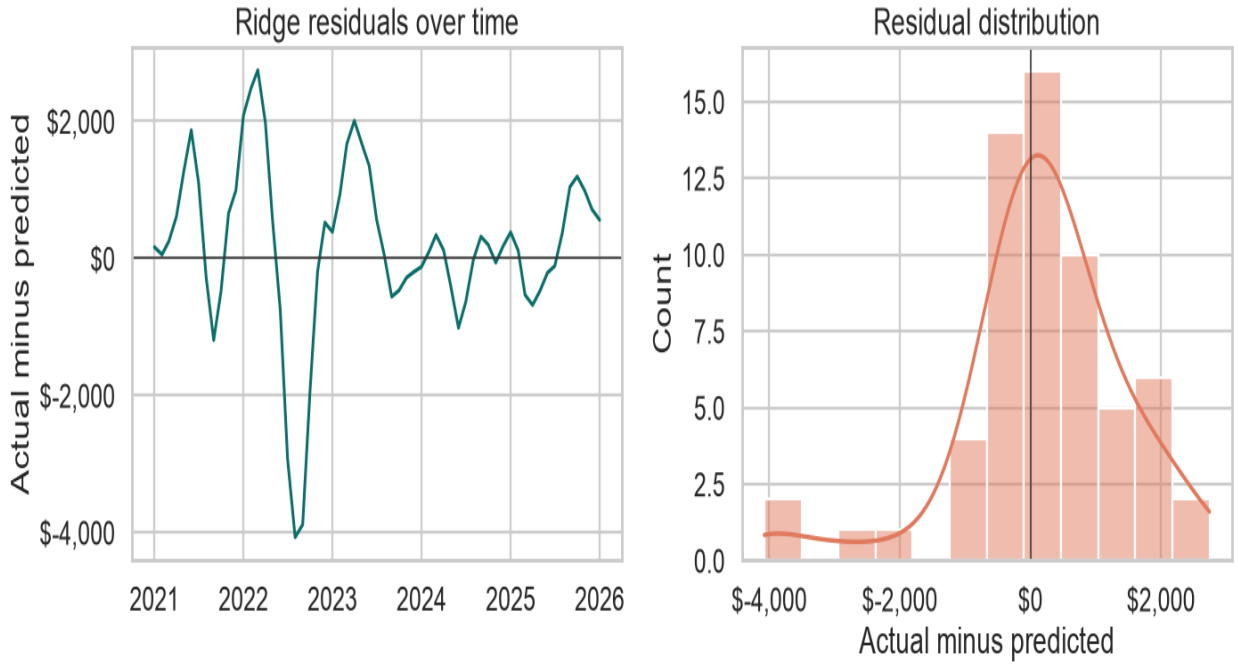


Figure 7. Residuals remain centered near zero, although errors cluster during periods of faster market change.

Why the tree models failed

Random forests and gradient boosting trees performed poorly after 2020 because much of the holdout series exceeded the training target range. Standard tree ensembles partition observed values; they do not naturally extend a rising linear level beyond the training range. A trend-aware transformation or direct modeling of monthly changes would be more appropriate if a tree model were required.

Limits on the conclusion

- The strong one-month fit does not demonstrate accurate long-range forecasting.
- The holdout includes a rapid housing expansion and cooling period, but one historical episode cannot represent every future market regime.
- Past ZHVI alone cannot explain why prices change or whether a market offers an attractive risk-adjusted return.

Recommendations and next steps

1. Operational forecast

Use the ridge pipeline as a one-month monitoring forecast. Refresh the model monthly, save each vintage, and compare live errors with both the fitted model and the persistence baseline.

2. Model the business question directly

For an investment-ranking project, change the target from next month's price level to a longer-horizon return, rent yield, or risk-adjusted appreciation measure. Use metro or county geography because state averages hide local variation.

3. Add explanatory market data

Combine ZHVI with Zillow rent and inventory measures, Census income and migration data, and mortgage-rate series. These inputs would help distinguish momentum from affordability and demand.

Conclusion

The final model provides an accurate short-horizon Florida ZHVI forecast under a leakage-safe evaluation. The larger lesson is equally important: a precise forecast of one index does not answer the full investment question. The next version should align the target and features with valuation, income, and risk.

Sources

Zillow Research. Housing Data. https://www.zillow.com/research/data/

Zillow Group. Real Estate Metrics public data documentation. https://www.zillowgroup.com/developers/api/public-data/real-estate-metrics/

Scikit-learn documentation. TimeSeriesSplit, Ridge, RandomForestRegressor, and GradientBoostingRegressor. https://scikit-learn.org/stable/

Course-project dataset snapshot and analysis notebooks included in this repository.